
What is Critical Metascience and Why is it Important?

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Abstract

Metascience uses a scientific approach to understand and improve science. *Critical metascience* takes a step back to question metascience's commonly-accepted assumptions, methods, problems, and solutions. In this article, I define critical metascience and explain how it can help to identify and address collective biases within metascience. I also highlight some emerging themes in the field. Finally, I consider some potential relationships between metascience and critical metascience and argue that the most functional model is one in which two fields are partially overlapping. I conclude that there needs to be greater recognition and dialogue between metascience and critical metascience in order to improve metascience's objectivity and approach.

Keywords: metascience; open science; questionable research practices; replication crisis; science reform

What is Metascience?

Modern metascience took off in the 2010s in response to the replication crisis and concerns about research integrity, and it has been “exploding” in the last year (Tim Errington, quoted in “Metascience can improve science,” 2025). But what is metascience? There are several definitions (UK Metascience Unit, 2025, pp. 4-5), but I like Peterson and Panofsky's (2023) approach of locating metascience at the intersection of the science of science, open science, and methodological activism (see Figure 1).

Expanding on Peterson and Panofsky's (2023) approach, we can distinguish between three types of metascience. *Basic metascience* use a scientific approach to understand and critique the way we do science. *Applied metascience* designs and tests interventions to improve science. Finally, *activist metascience* is a more organized approach focused on reforming research culture by changing norms, incentives, and policies in line with open science principles (e.g., Grant et al., 2025; Nosek, 2019; Nosek et al., 2012). Importantly, it is this latter “activist reform movement” (Vazire & Nosek, 2023, p. 3) that dominates modern metascience, setting the agenda for both basic and applied metascience (Bastian, 2021; Nielsen & Qiu, 2018; Peterson & Panofsky, 2021, 2023).

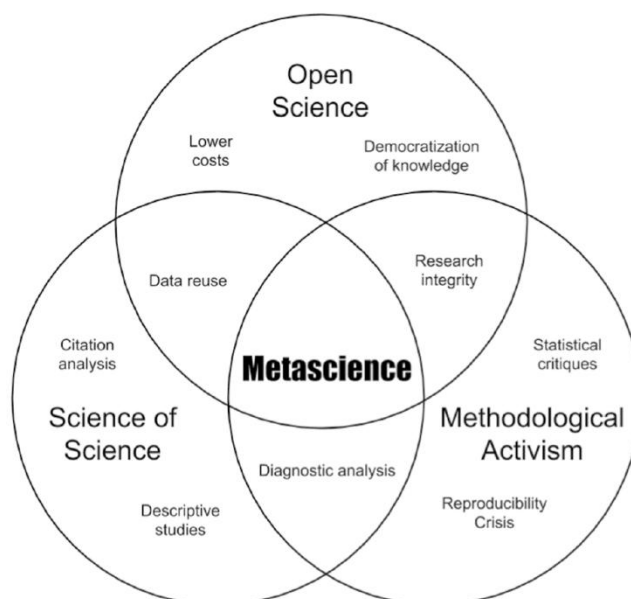


Figure 1. Peterson and Panofsky's (2023) conceptualization of metascience.

What is Critical Metascience?

I would define critical metascience as a multi-disciplinary approach that takes a step back to question some of metascience's commonly-accepted assumptions, methods, problems, and solutions. For this reason, critical metascience has also been described as “meta-meta-science” (Quintana & Heathers, 2023), and I think that term works well because it captures the idea that, if science is the subject of metascience, then metascience is the subject of critical metascience (see Figure 2). However, the “critical” part of critical metascience means that we are not just passively studying metascience; we are also actively criticizing some of its key assumptions and making suggestions for improvements.

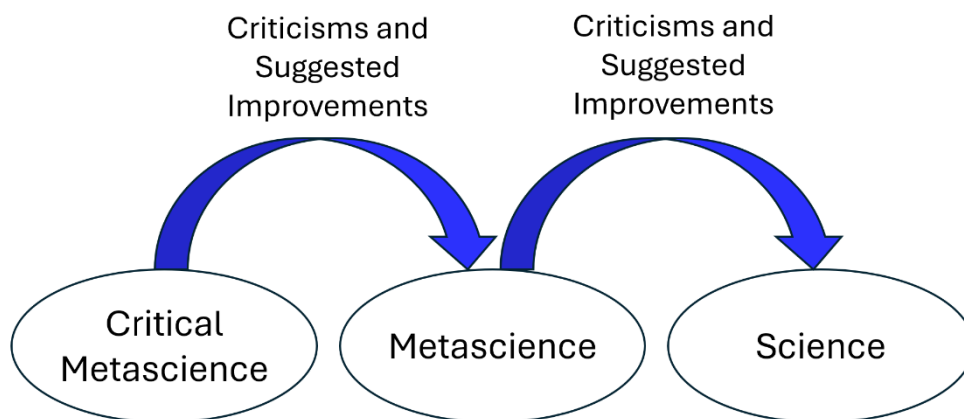


Figure 2. The relationship between critical metascience, metascience, and science.

So, in my view, “critical metascience” does not mean “metascientists who are critical” because metascientists are critical of themselves and others all the time. Instead, “critical metascience” refers to a special type of criticism that challenges metascience's dominant ideas and assumptions and offers new, countervailing perspectives. Hence, the word “critical” refers to disruptive, *paradigmatic criticism* (e.g., “Questionable research practices [QRPs] are not

necessarily problematic”) rather than more specific, *conventional criticism* that implicitly accepts the paradigm’s assumptions (e.g., “This measure of QRPs is not reliable”).

It is also worth noting that “critical metascience” does not necessarily imply a negative or contrarian stance. It can also provide constructive criticism. For example, although critical metascience may critique some science reforms, such as preregistration (e.g., Pham & Oh, 2021a, 2021b), it may also recommend *alternative reforms*, such as researcher reflexivity (e.g., Field & Pownall, 2025; Steltenpohl et al., 2023), as well as *alternative implementations of current reforms* (e.g., Bazzoli, 2022; Kessler et al., 2021).

Why is Critical Metascience Important?

Critical metascience is important because it can help to highlight and address collective biases in metascience. Of course, all scientific fields have biases. However, metascience’s biases may be particularly problematic because they are heavily influenced by its activist open science reform movement. As Hauswald (2021) explained:

If...the community of an entire research field is dominated by scholars with a particular activist affiliation...[and] if many or all members are biased in the same direction, their biases accumulate, as it were, to a collective bias. From a pluralist point of view, a dominance of persons with a particular activist background appears to be epistemically *detrimental* insofar as it reduces epistemic diversity. Research fields that are intimately connected to particular activist movements therefore seem to be particularly prone to such collective biases.

Hence, it is possible that the dominance of the activist reform movement may reduce epistemic diversity within metascience. This is not to dispute the good intentions of open science reformers. It is only to point out that the collective dominance of the assumptions behind those intentions may have a detrimental effect on the field’s epistemic approach (Bashiri et al., 2025, p. 9; Bazzoli, 2022; Rubin, 2023a; Rubin, 2023b, pp. 7-9).

Hauswald (2021) also predicts “the presence of various social mechanisms that produce or reinforce collective biases in such fields” (p. 612). And, sure enough, these sorts of “social mechanisms” have been observed in metascience (e.g., Bastian, 2021; Flis, 2022; Gervais, 2021; Hamlin, 2017, p. 692; Malich & Rehmann-Sutter, 2022, p. 265; Rubin, 2023b; Whitaker & Guest, 2020; Walkup, 2021). For example, Bastian (2021) noted that:

“Us and them” thinking and movement camaraderie lead some people to see criticism of fellow-travelers as “friendly fire” that’s out of line, leading them to keep criticisms to themselves. Advocates can also be very quick to jump into defensive attack mode instead of reflection when anyone criticizes. The combination of deterring incisive analysis and not engaging deeply with confronting thoughts is a shortcut to blinding yourself to movement and theory errors.

And this is where I think critical metascience can be helpful, because it deliberately steps back to reconsider some of metascience’s common assumptions, including implicit background assumptions that may be invisible from within the movement’s research culture (Kim et al., 2000, p. 68; Longino, 1990, p. 80; Popper, 1994, p. 135).

To be clear, I am not arguing that critical metascientists are *less* biased than metascientists. I am only arguing that they tend to have *different* and *more diverse* biases, and that it may be useful to take their perspectives into account in order to increase epistemic diversity (Hauswald, 2021; Hull, 1988; Longino, 1990; O'Rourke-Friel, 2025; Rubin, 2023a). Similarly, I am not arguing that activist metascience cannot improve science or that critical metascientists cannot also be activists. I am only arguing that activist metascience is susceptible to collective biases about what “science” is and how best to “improve” it, and that critical metascience can help to highlight and address these biases.

Questioning Some Common Metascience Assumptions

Figure 3 is a famous figure from Munafò et al.'s (2017) “Manifesto for Reproducible Science.” It provides a few examples of metascience’s core assumptions? (see also Chambers et al., 2014).

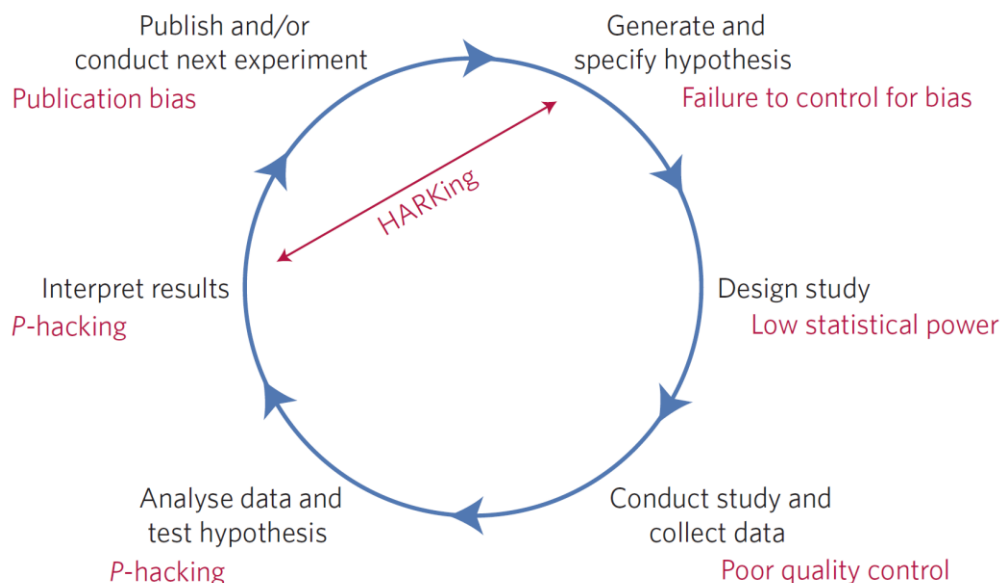


Figure 3. Munafò et al.'s (2017) proposed threats to the hypothetico-deductive method.

According to Munafò et al. (2017), researcher bias, low statistical power, HARKing, *p*-hacking, and publication bias lead to false positives and, consequently, low replication rates. Now, there is certainly good evidence and reasonable arguments to warrant each of these concerns. For example, metascientists and others often agree that:

1. **Exploratory results are more “tentative.”** E.g., Hardwicke & Wagenmakers (2023); Nelson et al. (2018); Nosek & Lakens (2014); Simmons et al. (2021b); Wagenmakers et al. (2012)
2. **QRPs are problematic.** E.g., Agnoli et al. (2017); Banks et al. (2016); Fraser et al. (2018); Hartgerink & Wicherts (2016); Miller et al. (2025); Pickett & Roche (2018)
3. **E.g., 1: HARKing is problematic.** E.g., Bishop (2019); Bosco et al. (2016); Kerr (1998); O'Boyle et al. (2014).

4. **E.g., 2: *p*-hacking is problematic.** E.g., Bishop (2019); Nagy et al. (2025); O’Boyle & Götz (2022); Reis & Frieze (2022); Simmons et al. (2011); Stefan & Schönbrodt (2023); Wicherts et al. (2016)
5. **Publication bias is problematic.** E.g., Bishop (2019); Chambers & Tzavella (2022); Errington et al. (2021); Schimmack (2020)
6. **Low replication rates are problematic.** E.g., Baker (2016); Open Science Collaboration (2015); Munafò et al. (2017); Pashler & Wagenmakers (2012); Schimmack (2020); Wingen et al. (2020)

These six claims are now so widely accepted within metascience that they are treated as almost indisputable facts! In this sense, they form part of an irrefutable Lakatosian “hard core” in the movement’s research program (Lakatos, 1978). Critically, however, Lakatosian research programs grow by qualifying their hard cores with auxiliary hypotheses that accommodate anomalies and make successful new predictions (e.g., “low replication rates are problematic in Situation X but not in Situation Y”; Lakatos, 1978, pp. 21, 50, 179). The problem is that this theoretical progress appears to be lacking in metascience, suggesting that the field may be in a pseudoscientific state (Lakatos, 1978, pp. 33-34).

Metascience’s limited theoretical progress cannot be attributed to the absence of important empirical and theoretical discrepancies. For example, there is now good evidence and arguments that challenge the generality of each of the previous six claims and call for some nuance in their articulation (for a brief review, see Rubin, 2025a):

1. **Exploratory results are more “tentative.” *BUT SEE*:** Devezer et al. (2021); Feest & Devezer (2025); Rubin (2017a); Rubin & Donkin (2024); Szollosi & Donkin (2021)
2. **QRPs are problematic. *BUT SEE*:** Fiedler & Schwarz (2016); Linder & Farahbakhsh (2020); Moran et al. (2022, Table 6); Ravn & Sørensen (2021); Rubin (2023b); Sacco et al. (2019); Ulrich & Miller (2020)
3. **E.g., 1: *HARKing* is problematic. *BUT SEE*:** Leung (2011); Oberauer & Lewandowsky (2019); Mohseni (2020); Rubin (2017b, 2022); Vancouver (2018)
4. **E.g., 2: *p*-hacking is problematic. *BUT SEE*:** Adda et al. (2020); Erasmus (2024); Gupta & Bosco (2023); Hartgerink (2017); Rooprai et al. (2023); Rubin (2024); Stanley et al. (2018); Ulrich & Miller (2020); Wegener et al. (2024)
5. **Publication bias is problematic. *BUT SEE*:** Brodeur et al. (2023); Dalton et al. (2012); Linden et al. (2024); Mathur & VanderWeele (2021); Ramos (2025); Van Aert et al. (2019)
6. **Low replication rates are problematic. *BUT SEE*:** De Boeck & Jeon (2018); Devezer et al. (2019, 2021); Fanelli (2018); Feest (2019); Firestein (2012, 2016); Greenfield (2017); Guttinger (2020); Haig (2022); Iso-Ahola (2020); Leonelli (2018); Lewandowsky & Oberauer (2020); Pethick et al. (2025); Rubin (2023b, 2025c); Stroebe & Strack (2014)

Isn't this work just standard self-critical metascience in action? Perhaps it should be, and maybe some of it is! However, if this sort of work was genuinely accepted as standard metascience, then it would be met with less defensiveness (Bastian, 2021; Flis, 2022; Gervais, 2021; Hamlin, 2017; Malich & Rehmann-Sutter, 2022, p. 265; Walkup, 2021), and its conclusions would be better integrated into the field to produce more complex and nuanced models of science (Greenwald et al., 1986, pp. 223-224; Lakatos, 1978, p. 50; Russell, 1912, p. 8). Instead, with few exceptions, "critical" work like this tends to be less acknowledged than standard metascience (Flis, 2022; Horbach et al., 2024; Rubin, 2023b), and relatively simplistic models endure (Flis, 2019, p. 170; Peterson & Panofsky, 2023, p. 169; Pham & Oh, 2021b, p. 168). This is a shame because it is precisely this type of work that we should be paying attention to if we want to tailor the right kinds of reforms to the right areas and implement them in the right ways.

Hence, in practice, "critical" work that questions common metascience assumptions is not usually treated as standard metascience. Indeed, its calls for caution, caveats, and nuance may be viewed as challenging metascience's more universalist claims of science-wide "crises" (Jamieson, 2018; Penders, 2024, p. 3).

Why Does Metascience Need Critical Metascience?

Why does metascience need a special, separate, "critical" field? What makes it different from other disciplines in this respect? Well, in fact, some other disciplines also have critical fields. For example, psychology has *critical psychology* (Parker, 2007), and, more recently, the area of *critical data studies* has arisen to question dominant views in data science (Iliadis & Russo, 2016).¹ But there are some unique aspects of metascience that make critical metascience particularly useful.

First, modern metascience arose in response to a "crisis" about replication and research integrity (Nielsen & Qiu, 2018; Pashler & Harris, 2012; Schooler, 2014). The problem is that, in a crisis, people often respond with a biased cognitive style. For example, they might experience a range of negative moral emotions, have a narrow focus on specific issues, submit to autocratic decision-making, and engage in consensus-seeking and groupthink (e.g., Antonetti et al., 2025; Paulus et al., 2022; Turner & Pratkanis, 1998). So, it is helpful to get the views of people who are not operating in this crisis mode.

Second, metascience is heavily influenced by psychologists because psychology was ground zero for the replication crisis (Open Science Collaboration, 2015). But this influence may give the field a somewhat unrepresentative view of science and its problems (Flis, 2019; Malich & Rehmann-Sutter, 2022; Moody et al., 2022; Morawski, 2022). Consequently, as Gervais (2021) argued, "a movement that has emerged from critical reflection on psychological science should be open to critical self-reflection on its own workings and open to wisdom and critiques from other fields that may have important theoretical insights" (p. 828; see also Hamlin, 2017, p. 691). Critical metascience's multi-disciplinary, paradigmatic criticism can help to provide these insights.

Third, activist metascience can be construed as a moral crusade to save science from self-interested scientists operating in a dysfunctional incentive system (Bastian, 2021; Knibbe et al., 2025; Morawski, 2020; Nosek et al., 2012; Penders, 2024; Peterson & Panofsky, 2023; Robinson-Greene, 2017). According to this view, activist metascientists may sometimes experience a "white hat bias" (Lilienfeld, 2020, p. 2) and feel morally justified in summarily ignoring or dismissing criticisms of science reforms. Consequently, as Bastian (2021) explained, "we need a broad approach to risk of bias in metascience, that encompasses our high risk of confirmation bias and ideological thinking."

Finally, and more generally, it is important to appreciate that critical metascience has the same rationale as metascience; it is just pitched at a higher level of abstraction (i.e., it is “meta-meta-science”). Hence, it would be inconsistent to argue that we need a special field to criticize and improve the way we do science (i.e., metascience), but that we do not need a special field to criticize and improve the way we do metascience (i.e., critical metascience)!²

Some Emerging Themes in Critical Metascience

Unlike metascience, critical metascience does not have a coherent narrative (e.g., QRPs ➡ false positives ➡ replication crisis). Nonetheless, we can identify some emerging themes in the area.

One line of work has questioned the meaning and interpretation of replication failures, arguing that low replication rates may not (a) represent a “crisis” and (b) occur throughout science (e.g., Fanelli, 2018; Firestein, 2016; Haig, 2022; Lewandowsky & Oberauer, 2020; Pethick et al., 2025; Stroebe & Strack, 2014).

Another line of work has argued that metascience is too heavily focused on replication, statistics, and methods, and that this preoccupation may encourage *naïve empiricism*, *statisticism*, and *methodologism* respectively (e.g., Haig, 2022; Proulx & Morey, 2021; Rubin, 2023b; van Rooij & Baggio, 2021). This sort of work often calls for a greater focus on theory and theory development (e.g., Dames et al., 2024; Flis, 2022; Lavelle, 2023; Szollosi & Donkin, 2021).

There is also work questioning the rationale for and implementation of various open science reforms, such as preregistration and open data (e.g., Berberi & Roche, 2022; Hicks, 2023; Navarro, 2020; Prosser et al., 2023; Rubin, 2020). One concern here is that these research practices may not translate well across diverse scientific disciplines and paradigms (e.g., Khan et al., 2024; Lamb et al., 2024; Lash & Vandenbroucke, 2012; MacEachern et al., 2019; Pownall, 2024).

Finally, there has been a continuous stream of articles arguing that metascience has a biased view of science that tends to exclude minority group scientists and approaches (e.g., Bazzoli, 2022; Hamlin, 2017; Hostler, 2024; Leonelli, 2022; Whitaker & Guest, 2020; Ulpts, 2024). There is also a really interesting line of work that examines the sociology of the metascience movement, dealing with its roots in psychology and its scandalizing and moralizing tendencies (Flis, 2019; Penders, 2024; Peterson & Panofsky, 2021, 2023).

There are further examples of these and other lines of work in the “Articles by Topic” section of my list of 200+ critical metascience articles (Rubin, 2025b). Note that these articles are authored by scholars from a variety of disciplines, including statisticians, natural scientists, psychologists, political scientists, sociologists, ethnographers, science and technology studies scholars, philosophers of science, and, of course, metascientists! I am sure that many of these scholars do not identify as “critical metascientists,” and some may even actively reject this label, which is fine! What unites the contributions in this list is not a shared identity but a shared focus on questioning metascience’s commonly-accepted assumptions, approaches, problems, and solutions. Hence, just as you do not have to be a “metascientist” to do metascience (James Wilsdon, quoted in Stafford, 2025), you do not have to be a “critical metascientist” to do critical metascience!

What is the Relationship Between Metascience and Critical Metascience?

It is interesting to consider the relationships that might exist between metascience and critical metascience. Here, I consider four possibilities.

First, the two areas could be considered as being completely separate from one another (see Figure 4).

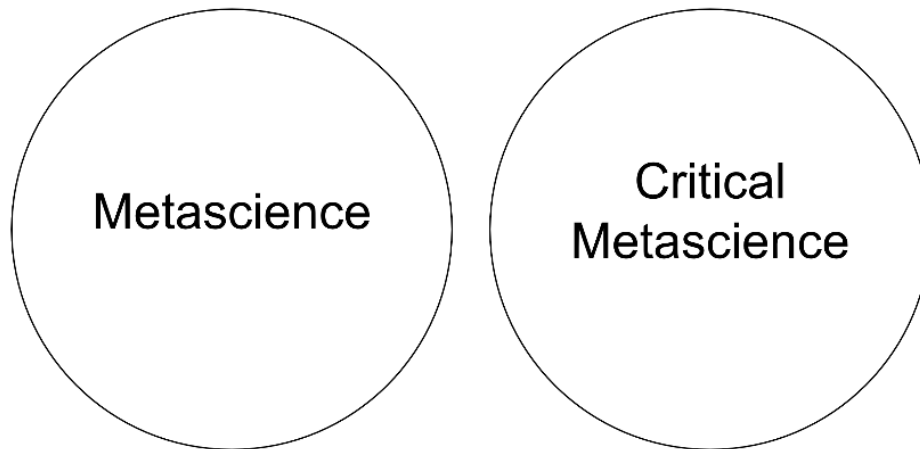


Figure 4. Metascience and critical metascience as two separate fields.

One advantage of this first model is that the clear separation between the two areas is likely to facilitate paradigmatic criticism. As Miller (2025) explained, “only an outsider, who adheres to a different conceptual framework, and isn’t committed to the insider’s standards, can thoroughly criticize the insider’s framework” (p. 44; see also Longino, 1990, p. 80). However, this model may also result in quite strong “us” and “them” relations that cause group polarization and intergroup bias. This model also fails to acknowledge that metascientists may sometimes engage in paradigmatic criticism (Syed, 2025). So, instead, we might consider a model in which there is some overlap between the two areas (see Figure 5).

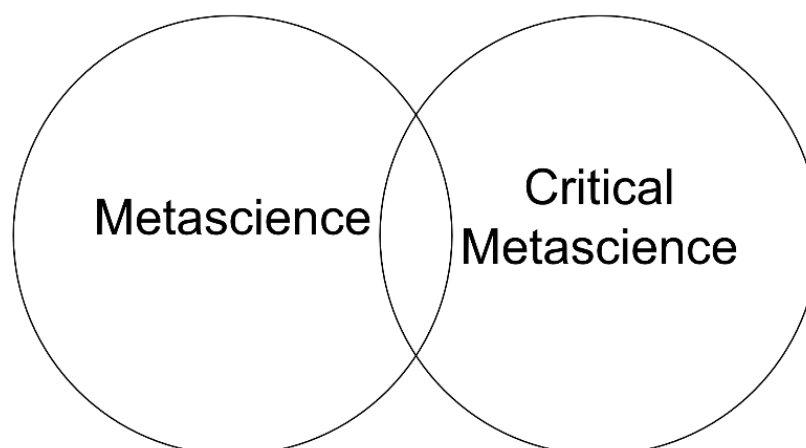


Figure 5. Metascience and critical metascience as partially overlapping fields.

In this second model, metascientists may sometimes step back to criticize some of their field's core assumptions, and critical metascientists may accept some of these core assumptions while criticizing others. This model continues to facilitate paradigmatic criticism, but it is less exclusionary and separationist than the first model and, hopefully, more likely to lead to productive interactions. Hence, it is my preferred model.

Another way to conceive this second model is as a continuum ranging from greater acceptance of metascience's core assumptions at one end to greater rejection of those assumptions at the other end (see Figure 6).

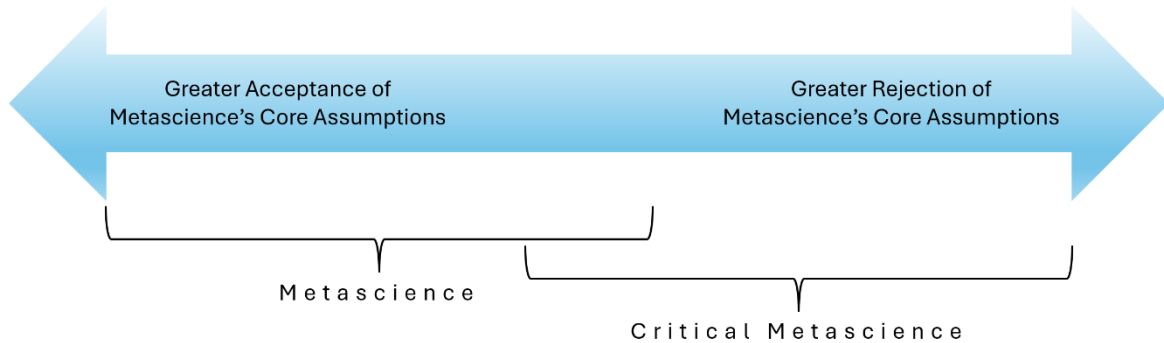


Figure 6. A critical metascience continuum.

In this case, more conventional criticism would occur towards the left end of the continuum, and more disruptive, paradigmatic criticism would occur towards the right end of the continuum. Again, this model is more flexible than the first because it allows the same people to occupy different points along the continuum at different times and for different projects.³

We can quickly consider two further models. First, we could view critical metascience as a subfield of metascience (see Figure 7).

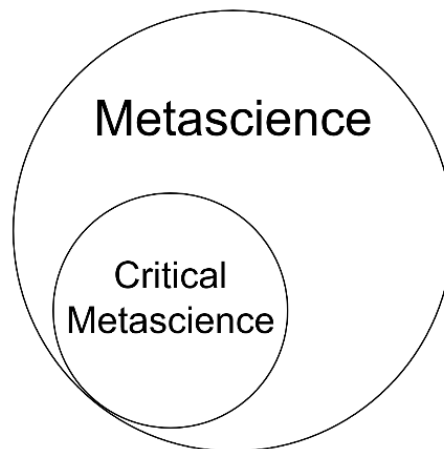


Figure 7. Critical metascience as a subfield of metascience.

Second, we could go further and consider critical metascience as being peppered into all aspects of metascience (see Figure 8).

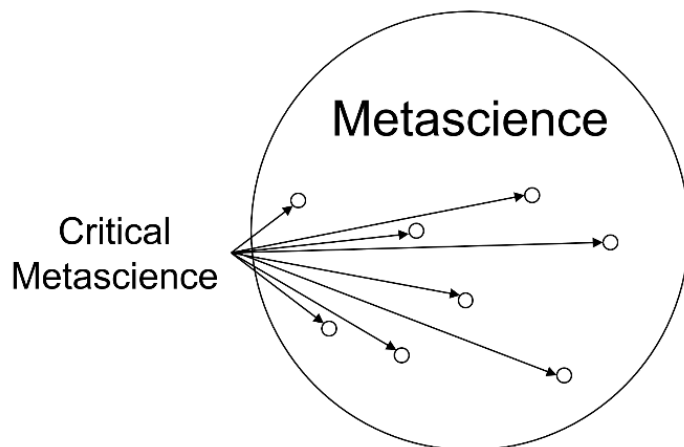


Figure 8. Critical metascience integrated throughout metascience.

The problem with these last two models is that the criticism comes from *within* metascience. Hence, it is more likely to be grounded in metascience's background assumptions and less likely to challenge those assumptions. Consequently, these last two models are less likely to generate the more radical, paradigmatic criticism that characterizes critical metascience. Indeed, it would be self-contradictory and epistemically incoherent for the majority of metascientists to routinely reject their own field's central assumptions, as implied in the last model.

What Needs to Change and Why?

To conclude, let me try to address the questions that were posed in a recent symposium on critical metascience (Ulpts & Bartscherer, 2025): What needs to change and why? I think one important change is for there to be greater recognition and dialogue between metascience and critical metascience.

For its part, metascience champions scientific criticism. For example, it advocates adversarial collaborations, red teams, effective peer review, error detection, and self-correction, and it encourage scientists to “bend over backwards to show how [they]’re maybe wrong” (Feynman, 1974). Consequently, it should welcome critical metascience as a useful addition to its own self-criticism toolkit. Of course, metascientists do not always need to concede to critical metascience arguments, but they should at least engage with those arguments publicly, formally, and carefully (Longino, 1990, p. 78; O’Rourke-Friel, 2025; Rubin, 2023b).

And, it takes two to tango! So, critical metascientists need to be prepared to reach out to metascientists in order to have productive exchanges rather than to simply criticize from afar (for a good example, see Pham & Oh’s [2021a, 2021b] discussion with Simmons et al. [2021a, 2021b] about preregistration). They should also be careful not to artificially homogenize metascience or criticize straw-person or outdated versions of metascience arguments (Field, 2022; Rubin, 2023b; Syed, 2025). Finally, they should be reflexive, continuously questioning their own perspectives, biases, and positionality (Bashiri et al., 2025, p. 7).

And why do we need this change? Well, I think that critical metascience can help to improve metascience by making it more objective. As Popper (1976) explained:

Scientific objectivity is based solely upon a critical tradition which, despite resistance, often makes it possible to criticize a dominant dogma. To put it another way, the objectivity of science is...[the result] of the friendly-hostile division of labour among scientists (p. 95)

Critical metascience can play an important part in Popper's "friendly-hostile division of labour" by offering a special type of criticism that exposes and challenges potential biases and dogma in metascience and by improving its objectivity and approach.

Epilogue: Comments from the Q&A Session at the 2025 Critical Metascience Symposium

This article is based on a presentation I gave in June 2025 as part of a Metascience 2025 Preconference Virtual Symposium convened by Sven Ulpts and Sheena Bartscherer. The symposium was titled "*Critical metascience: Does metascience need to change?*", and there is a recording of it [here](#).

The symposium ended with a really interesting Q&A session. One issue that came up was a concern that critical metascience arguments can be misappropriated by bad actors who want to sabotage science reform in order to further their anti-science agenda. I agree that this is a danger. However, this point should not be used as a justification for ignoring critical views. If we go down that road, then, to be consistent, we must also ignore activist metascientists' criticisms of the status quo because their narrative about the replication crisis and the need for urgent widespread reform can also be co-opted by bad actors for anti-science reasons (Biagioli & Pottage, 2022; Jamieson, 2018; Levy & Johns, 2016; O'Grady, 2025; Nosek, 2025). In summary, both critical and activist metascience arguments run the risk of misappropriation, but that does not imply that those arguments should be suppressed or ignored. It just means that we need to be *careful* about how we communicate our arguments in order to reduce the potential for their misuse.

More generally, as I mentioned earlier, critical metascience arguments are not necessarily anti-reform arguments. They may also be arguments for *alternative reforms* and/or *alternative implementations of current reforms*.

Finally, not every part of the scientific status quo may be problematic and in need of reform! Critical metascience can help to ensure we do not throw out the baby with the bathwater!

Endnotes

1. Critical psychology and critical data studies both adopt a critical theory approach that examines power structures, ideologies, and social injustices within their respective disciplines (Bohman, 2021). Although some critical metascientists may follow this approach, I view critical metascience as being broader than, and not constrained to, critical theory.
2. This last argument implies an infinite regress of ever more reflective metasciences (e.g., "meta-meta-meta-science")! However, each new meta-level will criticize a subject that becomes increasingly abstract and obscure. Hence, although I agree that critical metascience should be critiqued, I think we will quickly lose interest as we reach even higher levels of abstraction!
3. Consistent with Figure 6's continuum model, some of the articles in my list of critical metascience articles are less radical and disruptive than others.

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